

Problem 1:

Table

X_1	X_2	X_3	X_4	Y
0.56	0.60	0.38	0.50	Yes (1)
0.90	0.80	0.13	1.00	No (0)
1.00	0.60	0.25	0.33	Yes (1)
0.75	0.29	1.00	0.67	No (0)

(a) Find the cross entropy loss for the examples in the given Table using a Z score given by

$$Z = 0.5 X_1 + 0.25 X_2 + X_3 + 0.75 X_4 - 1.4$$

Assume you are using logistic regression for classification tasks.

(b) The initial Z score for a two input AND gate is $Z = X_1 + X_2 - 0.5$. Update the Z score through examining the training examples for only one cycle.

Problem 2:

Suppose, you are given the data below and you want to classify the labels in Y using the Logistic Regression Algorithm for Multiple Classes (Softmax Regression). Here, X_1 and X_2 are the features.

Instance	X_1	X_2	Y	Y_{encoded}		
				A	B	C
1	2.5	3	A	1	0	0
2	3	2.7	A	1	0	0
3	1.5	2	B	0	1	0
4	2	1.8	B	0	1	0
5	4	3.5	C	0	0	1
6	3.7	4	C	0	0	1
7	2.2	2.5	A	1	0	0
8	3.5	3.2	A	1	0	0

As there are two features (additionally, $X_0 = 1$) and three classes, the hypothesis function will look like below:

$$h_{\theta}(x) = \begin{bmatrix} P(y = A | x; \theta) \\ P(y = B | x; \theta) \\ P(y = C | x; \theta) \end{bmatrix}$$

$$\text{Where } P(y = i | x; \theta) = \hat{y}_i = \frac{e^{\theta_i^T \cdot x}}{\sum_{j \in \{A,B,C\}} e^{\theta_j^T \cdot x}}$$

Here, x is the feature vector (1×3 matrix) of a particular datapoint and θ_i is the parameter vector (3×1 matrix) specific to class i . In this equation, $\theta_{class}^T \cdot x$ is the dot product between θ_{class}^T and x ; not matrix multiplication between them.

Now, let's assume that after using Gradient Descent for 1000 iterations, you got the following matrix for θ :

$$\theta = \begin{bmatrix} \theta_A & \theta_B & \theta_C \end{bmatrix} = \begin{bmatrix} -0.10 & 4.42 & -5.32 \\ -0.14 & -1.36 & 1.23 \\ 0.18 & -0.92 & 0.23 \end{bmatrix}$$

- Your task is to find out the predicted class $\hat{y}$ for each data point in the given data. Show your calculations.
- What will be the updated parameter values after another iteration of Gradient Descent? [Consider, Learning Rate = 0.1]